

Vacuum Cleaner Robot – Corrected Python Program

Corrected Code

```
# Vacuum Cleaner Robot

# Initial room status
room = ["C", "D", "D", "D", "D", "C"]

print("Initial Room Status:")
for i in range(6):
    print(f"Tile {i + 1}: {room[i]}")

print("\nRobot Started Cleaning...\n")

# Robot checks each tile
for i in range(6):
    print(f"Robot is at Tile {i + 1}")

    if room[i] == "D":
        print("Tile is Dirty.")
        print("Cleaning Tile...")
        room[i] = "C"
        print("Tile Cleaned.")
    else:
        print("Tile is already Clean.")

    print("Moving to the next Tile.\n")

# Final room status
print("Cleaning Completed!\n")

print("Final Room Status:")
for i in range(6):
    print(f"Tile {i + 1}: {room[i]}")

print("\nAll tiles have been checked.")
```

Sample Output

```
Initial Room Status:
Tile 1: C
Tile 2: D
Tile 3: D
Tile 4: D
Tile 5: D
Tile 6: C

Robot Started Cleaning...

Robot is at Tile 1
Tile is already Clean.
Moving to the next Tile.

Robot is at Tile 2
Tile is Dirty.
Cleaning Tile...
Tile Cleaned.
Moving to the next Tile.

...

Cleaning Completed!
```

Final Room Status:

Tile 1: C

Tile 2: C

Tile 3: C

Tile 4: C

Tile 5: C

Tile 6: C

All tiles have been checked.